A NAS-Layer Inspection Tool for Verifying Rejection of Spoofed Emergency Registration in 5G Private Networks*

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Abstract

The 5G Emergency Registration procedure is an exceptional control-plane path in which the normal authentication flow may be shortened or omitted. If a core network fails to reject an unregistered User Equipment (UE) that abuses this path, unauthorized network access becomes possible, violating the "blocking unauthorized UEs" requirement of the Korea Internet & Security Agency (KISA) 5G private network security guideline. This paper proposes a Non-Access Stratum (NAS)-layer inspection tool that modifies the sendInitialRegistration() function of UERANSIM to send Emergency Registration requests from an unregistered International Mobile Subscriber Identity (IMSI), and judges the result from Access and Mobility Management Function (AMF) responses and UE state transitions. Experiments on an Open5GS testbed show that the AMF rejects the request with cause FIVEG_SERVICES_NOT_ALLOWED before the authentication procedure begins, confirming standard-compliant behavior.

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1 Introduction

5G private networks are increasingly deployed in smart factories and other mission-critical environments, where the core network handles User Equipment (UE) registration, authentication, and mobility management on behalf of the operator. To support exceptional situations, 3GPP TS 24.501 defines the Emergency Registration procedure [1], in which the normal authentication flow defined in TS 33.501 [2] may be shortened or skipped. If an unregistered UE abuses this exceptional path and the core network fails to reject it, unauthorized access becomes possible - a recurring source of authentication bypass in both 4G and 5G commercial networks [3][4]. Verifying compliance with the "blocking unauthorized UEs" requirement of the Korea Internet & Security Agency (KISA) 5G private network security guideline [5] therefore requires a practical inspection method, yet commercial UEs do not allow arbitrary manipulation of Non-Access Stratum (NAS) message fields. This paper proposes a NAS-layer inspection tool based on a modified UERANSIM [6], reproducing a spoofed Emergency Registration request from an unregistered UE and verifying whether an Open5GS core network rejects it before entering the authentication procedure.

2 Background

2.1 Emergency Registration Procedure

Registration establishes a UE's NAS connection with the Access and Mobility Management Function (AMF). 3GPP TS 24.501 defines five Registration types, including Emergency Registration, an exceptional type used when a UE that cannot complete normal registration still needs emergency services [1]. Because this path may proceed even without valid credentials, parts of authentication and security-context establishment mandatory for Initial Registration may be shortened or omitted [2].

2.2 Threat: Spoofed Emergency Registration

This relaxation makes Emergency Registration an attractive target. If an adversary controls a UE not provisioned in the operator's Unified Data Management (UDM) and sets the Registration Type to Emergency, the AMF must still reject the request [5]. Although exceptional NAS procedures have been repeatedly exploited for authentication bypass [3][4], no practical tool lets operators verify their core's enforcement. A noncompliant core may permit unauthorized access, enable IMSI enumeration, or allow PDU session establishment without valid credentials, violating [5].

3 Tools Design

The proposed tool modifies the NAS Mobility Management (MM) layer of UERANSIM [6] so that the Registration Type Information Element (IE) is always set to EMERGENCY_REGISTRATION, regardless of the emergency flag. The modified UE, using an IMSI not provisioned in the core network, sends this spoofed request via the UERANSIM next-generation NodeB (gNB) to the AMF. The tool judges the result as PASS when a Registration Reject is received and the UE remains in MM-DEREGISTERED before authentication begins.

4 Evaluation

The tool was evaluated on Ubuntu 22.04 LTS with Open5GS v2.7.7+ and UERANSIM v3.2.7. Executing the modified UE with an unregistered IMSI, the AMF returned Registration Reject (FIVEG_SERVICES_NOT_ALLOWED) before authentication began, and the UE transitioned to MM-DEREGISTERED. The result is judged as PASS, indicating that Open5GS complies with the "blocking unauthorized UEs" requirement of [5] for the Emergency Registration path.Conclusion

This paper proposed a UERANSIM-based inspection tool for verifying authentication bypass attempts in a 5G core network. The proposed tool changes the NAS message processing behavior of the UE simulator so that the UE sends Security Mode Complete without sending Authentication Response.

Experimental results showed that the Open5GS core network did not send Registration Accept or proceed to PDU Session Establishment for the abnormal UE. Instead, authentication failure and re-authentication procedures were observed. Therefore, the tested 5G core network properly blocked access from a UE that had not completed authentication. Future work will extend the tool to other NAS-based security inspection items, such as replay attacks and NAS integrity verification failures. It will also be applied to 5G core testbeds such as free5GC and OpenAir Interface for further validation

5 Conclusion

This paper proposed a NAS-layer inspection tool that reproduces spoofed Emergency Registration attempts from an unregistered UE by modifying UERANSIM, and judges core network compliance from the reject cause and UE state. Experiments on an Open5GS testbed confirmed rejection with cause before authentication. The tool enables systematic verification of the “blocking unauthorized UEs” requirement without modifying commercial UEs, supporting audits, regression testing, and KISA-based compliance workflows. Future work includes extending the tool to other NAS abuse scenarios (replay, identity spoofing, message tampering), and validating it on free5GC and OpenAirInterface.

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